Handout 8: The mystery powder

REMUS

Name the powder. Day 10, period 2. Team name: _____

An X-ray beam went through a powder and came out at the eight angles below (in degrees). Every crystal has its own fingerprint: the angles it always shows. The powder is a mix of **two** crystals from the five fingerprints below. Find which two.

The mystery pattern:

Peak	1	2	3	4	5	6	7	8
Angle	20.77	26.56	31.68	34.50	36.22	36.50	50.08	56.73

The same pattern drawn on a scale, each `^` one peak:

```
20                30                40                50                60                70
| . . . . | . . . . | . . . . | . . . . | . . . . | . . . . | . . . . | . . . . | . . . . |
  ^         ^         ^     ^  ^^                    ^           ^
```

The five fingerprints, each with its four strongest peaks, drawn on the same scale as the pattern, so read straight down: a mark lines up with a peak when the two sit within about a third of a degree of each other. The straight edge of a pencil or a card held upright helps.

```
quartz    20.9  26.6  36.5  50.1
  ^         ^                ^                        ^
```

```
zincite   31.8  34.4  36.3  56.6
                    ^    ^   ^                                  ^
```

```
rutile    27.4  36.1  41.2  54.3
               ^               ^         ^                        ^
```

```
halite    31.7  45.4  56.5  66.2
                    ^                           ^                    ^                ^
```

```
silicon   28.4  47.3  56.1  69.1
                  ^                                    ^               ^                       ^
```

Fingerprint	Peaks of the pattern it lines up with	Its marks that the pattern does not show
quartz		
zincite		
rutile		
halite		
silicon		

The two crystals in the powder: _____ and _____

Peaks of the pattern your two leave unexplained: _____

Which fingerprint looked promising at first and then let you down, and how did you catch it?

_____